

Title...

(Dated: June 2, 2026)

I. THEORETICAL REVIEW

A. Simplicial complex

Consider labels $i, j, k, \dots \in \{1, 2, \dots, N\}$. A simplicial complex is a set K of nodes identified by labels i , links identified by labels ij with $i < j$, triangles identified by labels ijk with $i < j < k$, and so on. More generally, the elements of K are sequences $i_0 i_1 \dots i_k$ called k -simplices that satisfy $0 \leq i_0 < i_1 < \dots < i_k \leq N$. In particular, a simplicial complex K satisfies the condition of topological closure, meaning that if $i_0 \dots i_k$ is a simplex in K , then $i_1 \dots i_{j-1} i_{j+1} \dots i_k$ is also a simplex of K for any $j = 0, 1, \dots, k$.

B. Chains

To each k -simplex $i_0 \dots i_k$ of a simplicial complex K we associate a vector $e_{i_0 \dots i_k}$. The vector space C generated by the linear combinations of the set $\{e_{i_0 \dots i_k} : i_0 \dots i_k \in K\}$ is called the space of chains. The set $\{e_{i_0 \dots i_k} : i_0 \dots i_k \in K\}$ denotes the canonical basis of C . Any chain $c \in C$ can be written as

$$c = \sum_{k=0}^{N-1} \sum_{i_0 \dots i_k \in K} c^{i_0 \dots i_k} e_{i_0 \dots i_k}$$

where the $c^{i_0 \dots i_k}$ are just scalar coefficients denoting the coordinates of c in the canonical basis.

C. Cochains

The dual space C^* of C is called the space of cochains. The elements of the dual basis are denoted by

$$e^{i_0 \dots i_k}$$

and satisfy

$$e^{i_0 \dots i_k}(e_{j_0 \dots j_k}) = \delta^{i_0}_{j_0} \dots \delta^{i_k}_{j_k}$$

Any cochain $f \in C^*$ can be written as

$$f = \sum_{k=0}^{N-1} \sum_{i_0 \dots i_k \in K} f_{i_0 \dots i_k} e^{i_0 \dots i_k}$$

where the $f_{i_0 \dots i_k}$ are scalar coefficients denoting the coordinates of f in the dual basis. Cochains can be thought as tensor fields defined on top of the simplicial complex K .

D. Boundary operator

The boundary operator $\partial : C \rightarrow C$ is a linear operator defined by its action on the basis vectors

$$\partial e_{i_0 \dots i_k} = \sum_{j=0}^k (-1)^j e_{i_0 \dots i_{j-1} i_{j+1} \dots i_k}$$

In other words, it maps a k -chain $e_{i_0 \dots i_k}$ to the alternating linear combinations of the $(k-1)$ -chains associated to the “boundary” of the simplex $i_0 \dots i_k$. It mimics the behavior of the boundary operator of a “smooth” topological space.

E. Coboundary operator

The coboundary operator $d : C^* \rightarrow C^*$ is the linear operator defined by the dual relation

$$(df)(c) = f(\partial c)$$

for any chain $c \in C$ and any cochain $f \in C^*$. It mimics the behavior of the differential operator over a “smooth” topological space.

The coboundary operator d mimics the role of the gradient operator in “smooth” settings.

F. Inner product

The space of chains can be equipped with an inner product

$$C^2 \ni c, c' \rightarrow \langle c, c' \rangle \in \mathbb{R}.$$

For any given and fixed $c \in C$, the mapping

$$C \ni c' \rightarrow \langle c, c' \rangle \in \mathbb{R}.$$

defines a covector or, in other words, a cochain denoted by $c^\flat$; i.e. an element of C^* . If the inner product is non-singular, the mapping $C \ni c \rightarrow c^\flat \in C^*$ is invertible. Its inverse is denoted by

$$C^* \ni f \rightarrow f_\sharp \in C.$$

This invertible mapping together with the inner product on C induces an inner product on C^* given by

$$C^* \ni f, f' \rightarrow \langle f, f' \rangle := \langle f_\sharp, f'_\sharp \rangle \in \mathbb{R}$$

for every $f, f' \in C^*$.

G. Dual boundary operator

The dual boundary operator $\partial^* : C \rightarrow C$ is the inner-product adjoint of the boundary operator ∂ . In other words, is defined by

$$\langle \partial^* c, c' \rangle := \langle c, \partial c' \rangle$$

for every $c, c' \in C$.

H. Dual coboundary operator

The dual coboundary operator $d^* : C^* \rightarrow C^*$ is the inner-product adjoint of the coboundary operator d . In other words, is defined by

$$\langle d^* f, f' \rangle := \langle f, df' \rangle$$

for every $f, f' \in C^*$.

The dual coboundary operator d^* mimics the role of the rotor operator in “smooth” settings.

I. Hodge Laplacian

The Hodge Laplacian is a generalization of the graph Laplacian to simplicial complexes. It is defined as the linear operator $L : C^* \rightarrow C^*$ given by

$$L = (d + d^*)^2 = dd^* + d^*d$$

since it can be shown that $d^2 = (d^*)^2 = 0$.

J. Hodge decomposition

Any cochain $f \in C^*$ admits a so called Hodge decomposition

$$f = g \oplus s \oplus h$$

where $g, s, h \in C^*$ are unique given f . In particular, the components g , s and h are called the gradient, solenoidal and harmonic components of f and are such that:

$$Lh = 0$$

i.e. h belongs to the kernel of the Hodge Laplacian L , there exists a $w \in C^*$ such that

$$g = dw$$

and, there exists a $u \in C^*$ such that

$$s = d^*u.$$

II. DESCRIPTION OF THE EXPERIMENTS

In our numerical experiments we start with an empirically obtained cochain f defined on the links, i.e.

$$f = \sum_{ij \in K} f_{ij} e^{ij}.$$

Then, we compute its Hodge decomposition to obtain the gradient component

$$g = \sum_{ij \in K} g_{ij} e^{ij},$$

the solenoidal component

$$s = \sum_{ij \in K} s_{ij} e^{ij}$$

and the harmonic component

$$h = \sum_{ij \in K} h_{ij} e^{ij}.$$

Notice, these are also quantities defined over the links (i.e., they look like vector potentials). The method also provides a “scalar potential” or potential over the nodes i :

$$w = \sum_i w_i e^i$$

whose “gradient” produces the gradient component

$$g = dw$$

of f . Crucially, modula an arbitrary constant that is chosen such that $w_i \geq 0$ together with the condition that the smallest rating is 0, the components w_i provide the ratings of the nodes. Additionally, the method provides a “tensor potential” or potential over the triangles ijk :

$$u = \sum_{ijk} u_{ijk} e^{ijk}$$

whose “rotor” produces the solenoidal component

$$s = d^*u$$

of f .

III. DISCUSSION

The ratings w_i in w inferred from f are basically determined by g via the equation $g = dw$ previously mentioned. Hence, the part $f - g = s \oplus h$ of f that is not explained by g somehow determines how “inconsistent” are the inferred ratings. In particular, $h = 0$ for a fully connected network. As a consequence, we can say that the ratio

$$\frac{|s|}{|f|} \in [0, 1]$$

somehow quantifies how “inconsistent” are the inferred ratings. If the ratio is zero, the ratings are fully consistent. If the ratio is one, the ratings make no sense. Just to be clear, here,

$$|s|^2 = \sum_{ij} |s_{ij}|^2$$

and

$$|f|^2 = \sum_{ij} |f_{ij}|^2,$$

i.e. these are the standard euclidean norms.

IV. DESCRIPTION OF THE DATA OBTAINED FROM THE EXPERIMENTS

In the folder:

`ranking-futbol/juan/experiments/2026-06-01/txt/`

we provide text files such as:

- `LIGA-METRIC-hodge-decomposition-nodes.txt`
- `LIGA-METRIC-hodge-decomposition-links.txt`
- `LIGA-METRIC-hodge-decomposition-triangles.txt`
- `LIGA-METRIC-hodge-decomposition-globals.txt`

where LIGA and METRIC represent any of the leagues and metrics.

- The first file:

`LIGA-METRIC-hodge-decomposition-nodes.txt`

contains the columns:

`i w[i]=w_i r[i]=r_i tr[i]=tr_i`

denoting the label or number associated to each node i (which it is also the true ranking), the corresponding inferred rating w_i , the corresponding inferred ranking r_i and the corresponding true rating tr_i from the final position in the table (not from the metrics).

- The second file:

`LIGA-METRIC-hodge-decomposition-links.txt`

contains the columns:

`l i j f[l]=f_ij g[l]=g_ij s[l]=s_ij h[l]=h_ij`

denoting the label or number l associated to each link, the corresponding nodes i and j , and the corresponding components of f , g , s and h on such link.

- The third file:

LIGA-METRIC-hodge-decomposition-triangles.txt

contains the columns:

$t \ i \ j \ k \ u[t]=u_{ijk}$

denoting the label or number t associated to each triangle, the corresponding nodes i , j and k , and the corresponding components of the u .

- The fourth file:

LIGA-METRIC-hodge-decomposition-globals.txt

contains the columns:

$|f| \ |g| \ |s| \ |h| \ |u|$

denoting the norm of the corresponding vectors in C^* . These are global characteristics.
